PROJECT CODE: NEOSMITH

Product Requirement Document (PRD)

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1. Executive Summary & Product Strategy

1.1 Product Vision & Brand Philosophy

NeoSmith is an enterprise-grade AI Organizational Intelligence Agent operating as a strategic Digital Twin co-pilot for the Chief Human Resources Officer (CHRO) and executive leadership. The product is built around a unique architectural concept—the cognitive fusion of two distinct operational mechanics inspired by *The Matrix*:

- **The Agent Smith Layer (Systemic & Structural Awareness):** Implements an omnipresent view of the entire corporate enterprise. It rejects traditional administrative silos, treating the company instead as a mathematically balanced, unified grid of nodes, skills, dependencies, and communication network weights.
- **The Neo Layer (Human-Centric Optimization):** Focuses on purpose-driven human potential enhancement. It actively engineers cross-departmental collaboration, breaks down operational bottlenecks, and builds corporate "human circularity" to repurpose talent dynamically, eliminating the costly organizational disruption of mass layoffs during strategic market pivots.

1.2 The Problem Statement

Traditional Human Capital Management (HCM) software architectures (e.g., SAP SuccessFactors, Workday) operate on rigid, slow-moving hierarchical relational databases. These systems record corporate structures as static reporting line trees. They completely fail to capture:

- **Informal Collaboration Networks:** The actual pathways through which work, information, and influence travel across cross-functional boundaries.
- **Dynamic Competency Mapping:** The non-linear relationship between adjacent skill sets and how easily an employee with one capability can pivot to satisfy a vacancy in another domain.
- **Cross-Departmental Talent Fluidity:** The friction-heavy process of internal mobility that routinely forces companies to lay off underutilized workers in one branch while spending premium resources recruiting external candidates for another.

1.3 Strategic Business Value & Investment Justification

NeoSmith introduces a new standard for human capital optimization:

- **Mitigation of Restructuring Friction:** By calculating explicit reskilling timelines based on adjacent competency paths, the system replaces blind headcount cuts with optimal resource shifting.

- **Operational Silo Breakdown:** Tracing and expanding informal interaction paths protects organizations from tribal knowledge capture and cross-departmental communication failures.
- **Retention of Strategic Competencies:** Identifying non-hierarchical network drivers ("Quiet Forces") prevents companies from losing mission-critical informal influencers during standard corporate restructurings.

2. Core Schema & Graph Architecture

To handle systemic enterprise modeling, NeoSmith abandons relational tables in favor of an interconnected Neo4j Graph Database schema. This data architecture establishes explicit relational integrity across 5,000 active operational nodes and over 16,000 cross-entity relationships.

2.1 Node Blueprint Definitions

> `:Employee`
> - `id` (String, Unique Primary Key): System-generated deterministic corporate identifier (email).
> - `name` (String): Full professional name.
> - `department` (String): Mapped to one of 10 primary corporate divisions (Marketing, Sales, Engineering, IT, Risk & Compliance, Finance, R&D, People, Operations, Logistics).
> - `influence_score` (Float): Value between 0.01 and 0.99. Modeled via a Gaussian distribution across standard staff, with a forced 5% injection of high-value non-managerial personnel (scores 0.75 to 0.90) representing hidden operational leaders.

> `:Role`
> - `id` (String, Unique Primary Key): Structural identifier alphanumeric string (e.g., ROLE_CEO, ROLE_IT_MANAGER).
> - `title` (String): Official standard job title.
> - `level` (String): Hierarchical band label (Junior, Senior, Specialist, Manager, Director, C-Level).

> `:Skill`
> - `id` (String, Unique Primary Key): Competency catalog unique identifier code.
> - `skill_name` (String): Explicit industry capability moniker (e.g., 'EV Powertrain Engineering', 'Python').
> - `category` (String): Domain bucket classification matching the primary corporate units plus Soft Skills and Hybrid categories.

> `:OpenPosition`
> - `id` (String, Unique Primary Key): Vacancy authorization serial index identifier.
> - `department` (String): Destination corporate organizational bucket.
> - `priority` (String): Operational critical modifier metric ('High', 'Medium', 'Low').

2.2 Relational Edges

- **`[:REPORTS_TO]`**
  Source: `:Employee` or `:OpenPosition` → Target: `:Employee` (Manager).
  Semantic Role: Enforces strict formal hierarchy constraints and supervisor approval lines.
- **`[:COLLABORATES_WITH]`**
  Source: `:Employee` → Target: `:Employee`.
  Properties: `weight` (Float, 0.25 to 0.95) representing transaction velocity and communication frequency.
  Affinity Constraints: Generation logic locks an 80% intra-departmental density threshold and a 20% cross-departmental bridge allocation to simulate natural corporate communication clusters.
- **`[:HAS_SKILL]`**
  Source: `:Employee` → Target: `:Skill`.
  Properties: `proficiency` (Integer, 2 to 5).
- **`[:HOLDS_ROLE]`**
  Source: `:Employee` → Target: `:Role`.
- **`[:FOR_ROLE]`**
  Source: `:OpenPosition` → Target: `:Role`.
- **`[:REQUIRES_SKILL]`**
  Source: `:Role` → Target: `:Skill`.
- **`[:EQUIVALENT_TO]`**
  Source: `:Skill` → Target: `:Skill`.
  Properties: `similarity` (Float), `reskill_time_weeks` (Integer). Defines training distance for non-linear talent mapping.

3. Product Mechanics & Agent Configuration

NeoSmith executes actions through a highly restricted, context-aware routing architecture. Instead of allowing unconstrained text-to-code generation, the agent is bound by semantic guardrails and optimized database interaction tools.

3.1 Cognitive System Guardrails

- **Contextual Mobility Protocol:** If an executive prompts the system regarding restructuring or workforce reallocation, the agent **must not** force the input of both a source and a target department. If the target destination is missing, the parameter passes as empty, triggering a global graph sweep to automatically surface placement options across all remote business branches.
- **Anti-Typo Semantic Gateway:** The agent is restricted from pushing colloquial or unverified text strings straight into graph queries. If a user queries vague phrases (e.g., "IT spots"), the agent must first route inputs through vector similarity search indices (`text-embedding-ada-002`) to resolve and lock onto verified `:Role` or `:Skill` nodes before continuing structural execution.

- **Multiverse Homonyms Defense:** The system stores duplicated common names across diverse corporate units. If a query results in multiple records, the agent must halt execution, present the conflict, and prompt the user to clarify unique identifiers or organizational coordinates.

3.2 Agent Tool Architecture

- **Tool 1: `find_skill_by_natural_language`**
  Type: Similarity Vector Search. Purpose: Resolves vague text into concrete, official corporate competence nodes. Post-Processing Routine: Matches the discovered `:Skill` node back to qualified individuals, returning the top 10 profiles sorted by their informal `influence_score`.
- **Tool 2: `find_role_by_natural_language`**
  Type: Similarity Vector Search. Purpose: Maps imprecise job definitions to official database roles, surfacing open headcount vacancies and their reporting structures.
- **Tool 3: `uc_talent_circularity_and_reallocation`**
  Type: Parameterized Cypher Template. Purpose: Executes organizational talent redistribution. It parses individual arrays or broad scopes, maps candidate profiles against the global requirement matrix of open slots, filters by skill proximity, and returns candidates ordered by minimal training duration (`TrainingWeeksRequired` ASC) and baseline authority (`influence_score` DESC).
- **Tool 4: `uc_ona_collaboration_routing`**
  Type: Parameterized Cypher Template. Purpose: Runs shortest-path calculations between individual employee nodes to evaluate cross-functional silos, operational distance, and cross-departmental communication bridges.
- **Tool 5: `uc_general_organization_exploration`**
  Type: Controlled Text2Cypher Engine. Purpose: General administrative exploration (e.g., calculating structural tallies, evaluating direct leadership trees, and mapping baseline node statistics).

4. Operational Capabilities & High-Value Scenarios

The power of NeoSmith is demonstrated across 11 key operational scenarios, which define the product's validation suite:

Scenario ID / Name	Description & Operational Mechanism
Scenario 1 Open Positions & Skill Requirements	Resolves vague occupational titles via the vector gateway and queries downstream structural dependencies to output complete requisite capability matrices.
Scenario 2 Executive-Exempt IC Staff Count	Performs administrative property aggregation across global employee sets, explicitly isolating independent operational contributors by filtering out management, director, and executive-level nodes.
Scenario 3 Skill-Based Sourcing (Influence)	Locates subject matter experts across specific technical competencies, sorting profiles by informal influence scores rather than standard hierarchy.
Scenario 4 Cultural Project Stakeholders	Sweeps the entire organization to pull out the top 5 global organizational leaders inside the informal network to drive project alignment strategies.
Scenario 5 Shortest Path ONA Check	Traces communication links between individuals across distinct branches, calculating network hops to measure inter-departmental friction.
Scenario 6 Dynamic Cross-Dept Reallocation	Automates multi-person workforce transfers during unit restructurings, identifying available vacancies across the organization without target sector constraints.
Scenario 7 Structural Silo Detection	Identifies vulnerable nodes by filtering and sorting the bottom 10 employees by ascending degree centrality, exposing isolated personnel.
Scenario 8A Bulk Reallocation Matrix	Generates a full-scale transition plan for an entire department, highlighting high-potential placement options that require active reskilling periods (Reskilling > 0).
Scenario 8B Targeted Individual Mobility	Maps single employee profiles (e.g., *Jake Peralta*) against global job openings, evaluating alternative internal career trajectories based on skill adjacencies.
Scenario 9 Flight Risk Vacancy Mitigation	Performs critical workforce risk mitigation. When a high-influence asset is flagged for exit, it sweeps global vacancies to identify priority optimization paths.

Scenario ID / Name	Description & Operational Mechanism
Scenario 10 ONA Silo Bridge Analysis	Lists active cross-departmental connections and evaluates edge weights between distinct administrative groups, exposing single points of communication failure.
Scenario 11 Hidden Mentor Upskilling	Identifies top internal subject matter experts to act as mentors for cross-functional employee transitions, optimizing training curves.

5. Product Advantages & Competitive Edge

Feature Category	Traditional HR Platforms (Workday, SAP)	NeoSmith Agentic GraphRAG Solution
Organizational View	Static traditional reporting lines.	Dynamic informal interaction pathways and influence maps.
Internal Mobility	Manual keyword matches or self-reporting profile applications.	Algorithmic talent matching optimized by skill adjacency and network influence scores.
Restructuring Approach	Blind mass layoffs based on cost centers.	"Human Circularity" mapping to protect talent assets and reduce structural attrition.
Search Mechanics	Strict keyword queries prone to typographical failures.	Vector semantic gateway coupled with structural graph templates.
Strategic Insight	Backward-looking historical operational reports.	Real-time simulation of talent realocations, mentorship setups, and silo identification.

6. Technical Roadmap

- **Phase 1: Dynamic Vector Pipeline Expansion**
  Incorporate real-time vector indexing directly into node updates, enabling semantic processing to extend beyond predefined role titles to include changing employee skill profiles.
- **Phase 2: Autonomous Native Graph Analytics**
  Integrate graph algorithms (such as shortest path) straight into the database layer, replacing heuristic influence scores with real-time centrality calculations.
- **Phase 3: Conversational Scenario Simulations**
  Build a sandboxed interface where users can stage macro-economic workforce pivots, allowing executives to safely run and evaluate hypothetical internal reallocations before updating production systems.